

MATHEMATICS

9709/12

Paper 1 Pure Mathematics 1 (P1)

October/November 2015

1 hour 45 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 75.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of 4 printed pages.

- 1 Functions f and g are defined by

$$f : x \mapsto 3x + 2, \quad x \in \mathbb{R},$$

$$g : x \mapsto 4x - 12, \quad x \in \mathbb{R}.$$

Solve the equation $f^{-1}(x) = gf(x)$. [4]

- 2 In the expansion of $(x + 2k)^7$, where k is a non-zero constant, the coefficients of x^4 and x^5 are equal. Find the value of k . [4]

3

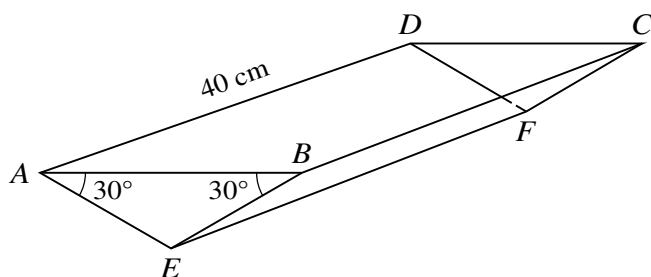


Fig. 1

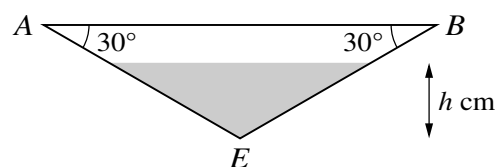
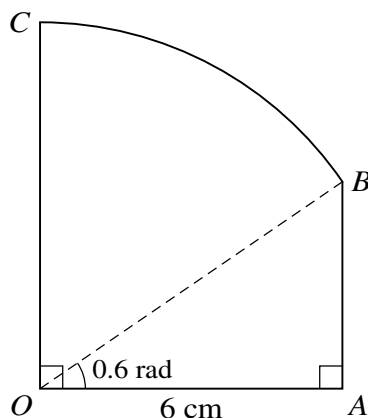


Fig. 2

Fig. 1 shows an open tank in the shape of a triangular prism. The vertical ends ABE and DCF are identical isosceles triangles. Angle $ABE = \text{angle } BAE = 30^\circ$. The length of AD is 40 cm. The tank is fixed in position with the open top $ABCD$ horizontal. Water is poured into the tank at a constant rate of $200 \text{ cm}^3 \text{ s}^{-1}$. The depth of water, t seconds after filling starts, is h cm (see Fig. 2).

- (i) Show that, when the depth of water in the tank is h cm, the volume, $V \text{ cm}^3$, of water in the tank is given by $V = (40\sqrt{3})h^2$. [3]
- (ii) Find the rate at which h is increasing when $h = 5$. [3]
- 4 (i) Prove the identity $\left(\frac{1}{\sin x} - \frac{1}{\tan x}\right)^2 \equiv \frac{1 - \cos x}{1 + \cos x}$. [4]
- (ii) Hence solve the equation $\left(\frac{1}{\sin x} - \frac{1}{\tan x}\right)^2 = \frac{2}{5}$ for $0 \leq x \leq 2\pi$. [3]

5



The diagram shows a metal plate $OABC$, consisting of a right-angled triangle OAB and a sector OBC of a circle with centre O . Angle $AOB = 0.6$ radians, $OA = 6$ cm and OA is perpendicular to OC .

(i) Show that the length of OB is 7.270 cm, correct to 3 decimal places. [1]

(ii) Find the perimeter of the metal plate. [3]

(iii) Find the area of the metal plate. [3]

6 Points A , B and C have coordinates $A(-3, 7)$, $B(5, 1)$ and $C(-1, k)$, where k is a constant.

(i) Given that $AB = BC$, calculate the possible values of k . [3]

The perpendicular bisector of AB intersects the x -axis at D .

(ii) Calculate the coordinates of D . [5]

7 Relative to an origin O , the position vectors of points A , B and C are given by

$$\overrightarrow{OA} = \begin{pmatrix} 0 \\ 2 \\ -3 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 2 \\ 5 \\ -2 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 3 \\ p \\ q \end{pmatrix}.$$

(i) In the case where ABC is a straight line, find the values of p and q . [4]

(ii) In the case where angle BAC is 90° , express q in terms of p . [2]

(iii) In the case where $p = 3$ and the lengths of AB and AC are equal, find the possible values of q . [3]

8 The function f is defined, for $x \in \mathbb{R}$, by $f : x \mapsto x^2 + ax + b$, where a and b are constants.

(i) In the case where $a = 6$ and $b = -8$, find the range of f . [3]

(ii) In the case where $a = 5$, the roots of the equation $f(x) = 0$ are k and $-2k$, where k is a constant. Find the values of b and k . [3]

(iii) Show that if the equation $f(x + a) = a$ has no real roots, then $a^2 < 4(b - a)$. [3]

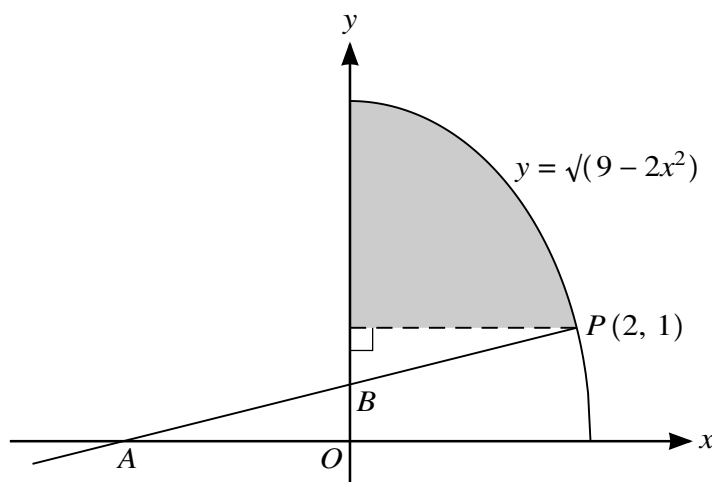
9 The curve $y = f(x)$ has a stationary point at $(2, 10)$ and it is given that $f''(x) = \frac{12}{x^3}$.

(i) Find $f(x)$. [6]

(ii) Find the coordinates of the other stationary point. [2]

(iii) Find the nature of each of the stationary points. [2]

10



The diagram shows part of the curve $y = \sqrt{9 - 2x^2}$. The point $P(2, 1)$ lies on the curve and the normal to the curve at P intersects the x -axis at A and the y -axis at B .

(i) Show that B is the mid-point of AP . [6]

The shaded region is bounded by the curve, the y -axis and the line $y = 1$.

(ii) Find, showing all necessary working, the exact volume obtained when the shaded region is rotated through 360° about the y -axis. [5]

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